**MYSchool ERP Master Blueprint** Canonical Technical Handover Dossier

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INSTITUTIONAL TECHNICAL BLUEPRINT

# MYSchool ERP Specification

Multi-Branch Architecture, High-Concurrency Workflows & Antigravity Operational Directives

🏛️ **Institution:** JSS School

⚡ **Engine:** Next.js 15+ & Supabase (Supavisor)

🚀 **Cache:** Upstash Redis (>90% Hit Ratio)

🔐 **Auth:** NextAuth.js (Auth.js v5)

🎯 **Concurrency:** 5,000+ Concurrent Requests

📌 **Status:** Client-Ready & Vibe-Coding Locked

## 01 Product Overview & Strategic Problem

*MYSchool PRD — Core Vision & Guardrails*

### 🎯 The Downside of Not Having This System

- **Delayed Duty-of-Care Alerts:** Manual 8:30 AM paper registers delay time-sensitive absence alerts to parents.
- **Accounting Ledger Drift:** Unverified digital fee receipts and bounced transfers cause ledger discrepancies.
- **Lost Statutory Records:** Uncoordinated departures risk the permanent loss of regulatory exit certificates.
- **Morning Roll-Call Crashes:** Unpooled serverless queries crash under 100+ concurrent classroom spikes.

### 🏁 What Success Looks Like

- **Digitizes, automates, and connects** all daily academic and administrative operations across JSS School.
- **100+ classroom teachers log 40 students** simultaneously in under 60 seconds with single-payload writes (~10ms).
- **Fee dues deduct only** after campus bursar verifies physical bank statement reconciliation.
- **Mandatory verified exit certificates** (TC, Character, No-Dues) strictly guard permanent student deletions.
- **Students retain lifetime access** to download official historical transcripts and report cards.

| ATTRIBUTE | SPECIFICATION FOR JSS SCHOOL |
|---|---|
| **Institutional Hierarchy** | Central Trust (JSS) → Physical Campuses (Main Campus, North Campus, East Campus) → Academic Wings (Primary, Middle, Secondary, Senior Secondary). |

| ATTRIBUTE | SPECIFICATION FOR JSS SCHOOL |
| --- | --- |
| **Primary Platform** | Responsive Full-Stack Web Application (Optimized for Desktop Office Terminals & Mobile Teacher Browsers). |
| **Core Tenancy Model** | Shared Multi-Branch PostgreSQL schema with dynamic URL scoping ( `/branches/[branchSlug]/[departmentCode]/...` ). |

# 02 Core Functional Modules & Workflows

*Seven Non-Negotiable Operational Pillars*

### ⚡ High-Speed Attendance Roll-Call

Teachers load class rosters in one click. Single-action submit dispatches one array payload (`createMany`) committing 40 records in ~10ms. Real-time webhook fires SMS/push alerts for unexcused absences.

### 💳 Manual Fee UTR Verification

Shows fee dues and UPI intents. Validates UTR uniqueness against duplicates and supports partial payments. **Balance is strictly untouched** until campus bursars audit physical bank statements.

### 📝 Student Leave Applications

Parents submit leave requests with date spans and doctor attachments, routed to Class Teachers. Approved leaves automatically flag calendar roll-calls as locked `EXCUSED_LEAVE`.

### 👨‍👩‍👦 Multi-Ward Single Parent Hub

Dedicated single-parent family model (no dual-parenting complexity). One login manages multiple wards across campuses with persistent top navbar switching.

### 🤖 Smart Substitution Assistant (Timetables)

Real-time collision engine detects teacher absences, queries free period slots in the same department wing, and generates 1-click ranked faculty matches without mutating the master timetable.

### ⚖️ Dual-Signoff Exam Grade Moderation

Subject teachers submit student marks as Draft; Wing Coordinators / HODs inspect class curves and averages before signoff. Grades publish to parents only upon final approval.

### 📊 Role-Adaptive Clean Dashboards

Zero-clutter interface tailoring: Teachers see attendance matrices and period schedules; Bursars see split-screen bank audits; Parents see 1-tap sibling switchers and fee cards.

### ⚡ Read-Heavy Caching via Redis (Upstash)

Master calendars, department topology, and subject lists cached at the edge (1h–24h TTL) with >90% hit ratio, offloading 80% of database load and protecting connection pool slots.

### 🔒 Exit-Certificate Deletion Barrier

Permanent student deletions require typed `admission_no` confirmation and are hard-blocked until 4 statutory certificates (TC, Character, Migration, No-Dues) are verified.

### 📄 Background Batch PDF Engine

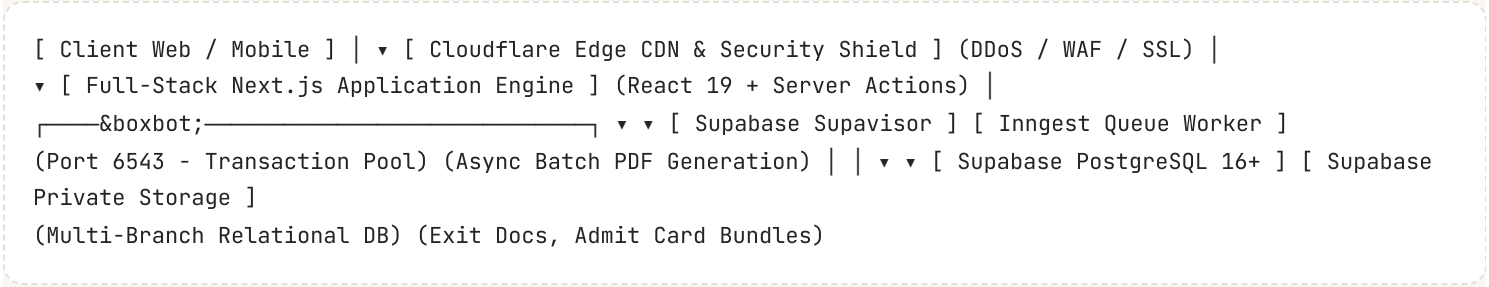
Buzzer Token pattern hands back a `jobId` in ~30ms. Inngest renders 2,000 admit cards and term receipts directly to Supabase Storage with 24-hour pre-signed download links.

### 📅 Academic Cycles & Lifetime Historical Transcripts

JSS School defines annual academic cycles. Past attendance archives to cold storage while **students retain permanent lifetime access to download official transcripts**. Staff attendance is tracked with daily timestamps.

# 03 System Architecture & Concurrency Strategy

*Engineering Stability for 5,000+ Users*




### 1. Supavisor Pooling

Solves serverless socket exhaustion. 500 concurrent teacher requests multiplex across **15–25 physical database connections** via transaction pooling on port 6543.

### 2. Direct Pre-Signed Uploads

Large student admission PDFs and exit certificates upload directly from client browsers to Supabase Storage, **bypassing Next.js server memory and bandwidth**.

### 3. Asynchronous PDF Jobs

Inngest handles mass admit cards and fee receipts off-thread. Administrators receive an active progress bar and a **24-hour signed download URL** upon job completion.

## TECHNOLOGY STACK & FRAMEWORK SPECIFICATIONS (7 TIERS)

| ARCHITECTURAL TIER | TECHNOLOGY & FRAMEWORK | TARGET / MODE | CORE ROLE & ARCHITECTURAL JUSTIFICATION |
|---|---|---|---|
| **Monorepo & Build** | Turborepo + pnpm | v2.4+ / pnpm v12 | Workspace management, cached builds, shared packages (shared, ui, api-client, config). |
| **Backend API** | NestJS (Fastify Adapter) | Node LTS / Fastify | Modular monolith API ( `main.ts` ); dependency injection, guards, DTOs, OpenAPI contract. |
| **Worker Process** | BullMQ with Redis | `worker.ts` | Off-thread background processing: Puppeteer PDF pool, email queue, imports, report batch runs. |

| ARCHITECTURAL TIER | TECHNOLOGY & FRAMEWORK | TARGET / MODE | CORE ROLE & ARCHITECTURAL JUSTIFICATION |
|---|---|---|---|
| **Frontend & UI** | React 18 + Vite | SPA / PWA | Client-rendered application with TanStack Query server-state caching and Role-Adaptive Dashboards. |
| **Language & Types** | TypeScript | v5.x+ (Strict Mode) | End-to-end type safety shared across API, web, and packages via `@campus/shared`. |
| **Styling & Design** | Tailwind CSS + shadcn/ui | Radix Primitives | Accessible institutional design tokens, high-density academic grids, and mobile-first tables. |
| **Relational Database** | PostgreSQL 16+ | RLS Enabled | ACID relational integrity, JSONB configuration, and strict tenant isolation via database policies. |
| **ORM & Access** | Prisma ORM | Dual-URL Pooler | Typed database client, versioned migrations, and transaction-scoped RLS setting. |
| **Cache & Locks** | Redis (ioredis) | Standalone / Cluster | Read-heavy master caching (>90% hit rate) and atomic payment idempotency locks ( `SET NX EX 60` ). |
| **Auth & Authorization** | JWT + CASL | `@casl/ability` | Stateless access tokens, rotating refresh cookies/body tokens, and fine-grained ABAC permissions. |
| **Object Storage** | S3-Compatible (R2 / S3) | S3 Buckets | 60-second HMAC presigned direct client uploads; 0MB server memory overhead during transfers. |
| **Document Compiler** | Puppeteer (Headless Chrome) | Worker Pool | High-fidelity HTML-to-PDF rendering of marksheets, admit cards, receipts, and certificates. |
| **Validation Layer** | Zod + class-validator | v3.24+ | Runtime DTO validation on API endpoints and client-side forms from shared contracts. |

## FAILURE HANDLING & CONCURRENCY RESILIENCE MATRIX

| FAILURE SCENARIO | ROOT CAUSE & REAL-WORLD RISK | TECHNICAL HANDLING STRATEGY & SAFEGUARD |
|---|---|---|
| **8:30 AM Attendance Surge** | **Serverless Socket Exhaustion:** 100+ teachers submitting simultaneously spins up hundreds of serverless threads, depleting | **Bulk Payload + Supavisor Pooling:** Single array payload of 40 students committed in ~10ms via Prisma `createMany`. Supavisor (Port 6543) transaction mode holds sockets only during |

| FAILURE SCENARIO | ROOT CAUSE & REAL-WORLD RISK | TECHNICAL HANDLING STRATEGY & SAFEGUARD |
|---|---|---|
| | PostgreSQL's 100-connection limit and crashing with `500 Internal Error`. | active write, multiplexing 500+ callers into **15–25 physical connections**. |
| **Batch PDF Generation** | **HTTP 504 Gateway Timeout:** Compiling 500+ admit cards with student photos and barcodes takes 1–3 minutes, exceeding the 60s reverse proxy / browser timeout limit. | **Inngest Async Token Pattern:** Offloaded to Inngest with instant 200 response and `jobId` in ~30ms. Headless worker renders in chunks, streams PDFs to Supabase Storage, and issues a 24h pre-signed download link. |
| **Invalid Fee UTR Number** | **Ledger Drift & Fraud:** Naive systems deduct student fee dues immediately on submission. If bank transfer fails or UTR is fake, accounting ledgers desynchronize. | **Two-Step Manual Audit Queue:** Submissions are tagged `PENDING_VERIFICATION`; **due balance remains untouched** until campus bursar audits bank records and clicks `[VERIFY & APPROVE]`. |
| **Unverified Student Deletion** | **Regulatory Liability & Lost Transcripts:** Deleting a student before statutory clearance violates education board mandates and destroys academic history. | **Statutory Exit Barrier Guard:** Hard programmatic check in deletion API. Verifies that Transfer Certificate (TC), Character Certificate, and No-Dues Slip exist with status `VERIFIED` before deletion. |
| **Heavy Document Uploads** | **Server Memory Exhaustion & Timeouts:** Uploading 15MB certificates through server threads exhausts Node.js RAM and triggers HTTP 504 timeouts on mobile. | **Presigned S3 Direct Uploads (`src/lib/storage.ts`):** Next.js signs a 60s HMAC token. Client browser streams file directly to S3 / Cloudflare R2, keeping server RAM at **0MB** and CPU at **0%**. |
| **Mobile Payment Double-Clicks** | **Duplicate Ledger Entries:** Impatient parents on flaky mobile connections double-tap "Submit", generating race conditions and duplicate UTR records. | **Strict Idempotency Engine (`src/lib/idempotency.ts`):** Client modal attaches a UUID. Server executes an atomic Redis lock (`SET key NX EX 60`), returning the cached pending response for subsequent taps. |

## 04 Enterprise Data Model & Entity Relations

*JSS School Relational Hierarchy Overview*

| ENTITY TABLE | KEY FIELDS & CONSTRAINTS | RELATIONAL PURPOSE |
|---|---|---|
| `organizations` | `id, name ("JSS School")` | Root institutional tier for group-wide executive leadership. |
| `branches` | `id, org_id, name, slug ("main", "north")` | Physical campuses with localized principals and staff. |
| `departments` | `id, branch_id, name, code ("PRIM", "PUC")` | Educational divisions mapped to distinct classes and HODs. |
| `users` | `id, branch_id, email, role, password_hash` | Role-based authentication across central and campus staff. |
| `students` | `id, user_id, class_id, admission_no, guardian_id` | Enrollee profile linked to guardian and academic class. |
| `fee_invoices` | `id, student_id, total_due, paid_amount, status` | Ledger tracking student fee schedules and verified balances. |
| `payment_submissions` | `id, invoice_id, utr_number, status, verified_by_id` | Manual transaction ID store audited by campus bursars. |
| `leave_applications` | `id, student_id, start_date, end_date, status` | Absence request store linked to class teacher approval loops. |
| `exit_certificates` | `id, student_id, certificate_type, file_url` | Mandatory barrier documents required before student purge. |

# 05 Antigravity AI Agent Instructions (AGENTS.md)

*Operational Laws for Autonomous Vibe Coding*

**1. Inspect** →

Examine existing schema & folder boundaries

**2. Plan** →

Identify affected types & verify non-destructive rules

**3. Implement** →

Write minimal, isolated code in feature directory

**4. Verify** ✓

Run compiler check (`tsc --noEmit`)

## ✓ DO WORKING RULES

- ✓ Scope all queries by `branchId` AND `departmentId` .
- ✓ Use single-array `createMany` inserts for morning roll-calls.
- ✓ Import database singleton from `@/lib/db` (Supavisor pool).
- ✓ Require verified exit certificates before student deletion actions.
- ✓ Separate domain logic into `features/*/server/*.service.ts` .

## ✖ AVOID STRICT PROHIBITIONS

- ✖ Never execute `prisma db push --force-reset` or drop columns.
- ✖ Never execute unconstrained school-wide database lookups.
- ✖ Never instantiate `new PrismaClient()` inside feature files.
- ✖ Never deduct fee balances automatically without staff verification.
- ✖ Never generate batch PDFs on the main Next.js HTTP thread.

## ⚙️ Core Engineering & Code Hygiene Directives

- **Feature-Sliced Architecture:** Organize code strictly under `src/features/[feature]/` (components, service, schemas).
- **Dumb UI, Smart Services:** UI components only handle render/local state; all business logic lives in `*.service.ts`.
- **Uniform API Envelopes:** Enforce `{ success, data, error, metadata }` across all Route Handlers and Server Actions.
- **Universal Error Handling:** Use custom error classes (`AppError`, `UnauthorizedError`) with safe try/catch envelopes.
- **Strict Naming Standards:** PascalCase for components, camelCase for functions/hooks, snake_case for PostgreSQL entities.
- **Zod Environment Validation:** Validate all runtime variables at boot via `@/config/env.ts`; never read `process.env` directly.
- **Proactive Dead-Code Removal:** Eliminate unused imports, dead functions, and obsolete code during every feature cycle.

## 📋 Pre-Release Validation Checklist

- ✓ TypeScript compiles with zero errors ( `tsc --noEmit` )
- ✓ Exit document barrier stops uncertified deletions
- ✓ Multi-branch context preserved across route changes
- ✓ UTR submission status stays PENDING until staff approval
- ✓ Attendance commits in single payload under 20ms
- ✓ Historical student report cards downloadable across years

## 06 Role-Based Access Control & Identity Matrix

*Seven Non-Negotiable Institutional Clearances*

### 🏛️ Super Admin (JSS Board)

Unrestricted trust-wide access; instant cross-campus switching; consolidated audit at `/all/all/analytics`; global fee policies.

### 🏫 Branch Admin (Principal)

Campus locked (`User.branchId`); campus admissions, staff substitution approvals, cycle promotion, and post-11 AM attendance override.

### 📚 Department Head (HOD)

Wing locked (`branchId + departmentId`); free teacher substitution assignment, wing leave approvals, and terminal grade book review.

### 👨‍🏫 Class Teacher (Faculty)

Classroom locked; submits 8:15–11:00 AM attendance register (single-payload `createMany`), reviews medical leaves, enters marks.

### 💼 Management (Bursar)

Bank statement UTR verification queue, fee ledger deductions, digital receipts, admission & exit certificate audit clearance.

### 👪 Parent / Guardian

Multi-ward family scoping; top navbar dropdown toggles active child context without re-auth; fee UTR payments, leave applications.

### 🎓 Student (Enrolled Pupil)

Personal profile view; timetable inspection, exam schedule view, fee dues, and **permanent lifetime access** to download official digital transcripts across all completed years.

| FUNCTIONAL CAPABILITY | SUPER ADMIN | PRINCIPAL | DEPT HEAD | CLASS TEACHER | MANAGEMENT | PARENT / STUDENT |
|---|---|---|---|---|---|---|
| **Cross-Branch Toggling** | Unrestricted | Campus Locked | Campus Locked | Campus Locked | Campus Locked | Ward Locked |
| **Class Attendance Submit** | View / Audit | View / Override | View / Override | Submit Assigned | None | View Own |
| **Attendance Lock Override** | Full Override | Permitted (>11 AM) | With Audit Note | None (Frozen) | None | None |
| **Fee Structure Config** | Global Policy | Campus Level | None | None | Create / Modify | None |

| FUNCTIONAL CAPABILITY | SUPER ADMIN | PRINCIPAL | DEPT HEAD | CLASS TEACHER | MANAGEMENT | PARENT / STUDENT |
|---|---|---|---|---|---|---|
| **UTR Verification & Ledger** | Full Audit | View Only | None | None | Approve / Reject | Submit UTR |
| **Student Leave Approvals** | Full Override | Campus Level | Department Level | Assigned Class | None | Submit Leave |
| **Batch Admit Card Printing** | Permitted | Permitted | Permitted | None | None | None |
| **Permanent Student Deletion** | Sign-off Only | Execute (Exit Cert) | Blocked | Blocked | Blocked | Blocked |
| **Lifetime Grade Downloads** | Full Trust | Campus Level | Department Level | Class Level | None | View / Download |

# 07 Architecture & Engineering Decision Records (ADRs)

*Comprehensive Log of 24 Meaningful Architectural & Functional Choices*

### 1. Technical Architecture (ARCH)

**Unified Next.js 15+ Monorepo:** Eliminates CORS, shares TypeScript/Zod types, and cuts hosting bills by 50%. Supavisor transaction pooler (:6543) stops 8:30 AM socket crashes. Edge JWT eliminates DB hits in middleware.

### 2. Functional Workflows (FUNC)

**Manual UTR Verification & Exit Barrier:** Zero automatic fee balance deductions; bursar audits bank slips before ledger cut. Programmatic guard blocks student purges until 4 statutory exit certificates are verified.

### 3. Reliability & Performance (NFR)

**Presigned S3 & Idempotency:** 60s direct client uploads bypass server RAM (0MB buffer). Redis distributed locks (`SET NX EX 60`) prevent double fee submissions on slow mobile connections.

| ID | DECISION TITLE | CATEGORY | STATUS | CORE ARCHITECTURAL RATIONALE & OPERATIONAL IMPACT |
|---|---|---|---|---|
| **ADR-001** | Git-Centric Live Web Docs | STRAT | Implemented | Immediate 1-click evaluation for employer via GitHub Pages; atomic Git version lockstep. |
| **ADR-002** | Multi-Format Dossier Suite | STRAT | Implemented | Fidelity across Web Portal, Printable Blueprint HTML, Headless PDF, and Offline Zip. |
| **ADR-003** | Unified Next.js 15+ Monorepo | ARCH | Implemented | Eliminates CORS, shares TypeScript/Zod schemas, and reduces infrastructure overhead by 50%. |
| **ADR-004** | URL-Scoped Tenancy Routing | ARCH | Implemented | Independent browser tab isolation ( `/[branch]/[dept]` ); deep-linking without cookie collision. |
| **ADR-005** | Managed Supabase PostgreSQL | ARCH | Implemented | ACID relational integrity; 7-day automated PITR; zero Linux VM maintenance. |
| **ADR-006** | Supavisor Pooling (:6543) | NFR | Implemented | Multiplexes 500+ callers into 15–25 physical sockets; stops 8:30 AM socket crashes. |

| ID | DECISION TITLE | CATEGORY | STATUS | CORE ARCHITECTURAL RATIONALE & OPERATIONAL IMPACT |
|---|---|---|---|---|
| **ADR-007** | Dual-Port DB Strategy | ARCH | Implemented | Port 6543 for pooled queries; Direct Port 5432 for automated Prisma migrations. |
| **ADR-008** | Prisma Singleton Client | ARCH | Implemented | Centralized singleton prevents connection pool leaks during Next.js hot reloads. |
| **ADR-009** | NextAuth v5 Edge JWT | ARCH | Implemented | 0ms DB overhead in Edge middleware; stateless verified role security across 7 roles. |
| **ADR-010** | Strict TypeScript & Zod | ARCH | Implemented | End-to-end compile-time and runtime validation on all mutations and env vars. |
| **ADR-011** | Bulk Attendance Array Payload | FUNC / NFR | Implemented | 40-student single array committed in ~10ms; async absence alert notification. |
| **ADR-012** | Two-Step Manual UTR Verification | FUNC | Implemented | Zero automatic balance deductions; bursar bank reconciliation prevents ledger drift. |
| **ADR-013** | Statutory Student Exit Barrier | FUNC | Implemented | Hard programmatic check requiring 4 verified clearance slips before student purge. |
| **ADR-014** | Single-Parent Multi-Ward Hub | FUNC / UX | Implemented | 1-click header sibling switcher; no duplicate parent accounts across grades/wings. |
| **ADR-015** | Smart Substitution Assistant | FUNC | Implemented | Timetable collision engine finds free teachers in same wing; 1-click duty assignment. |
| **ADR-016** | Dual-Signoff Grade Moderation | FUNC | Implemented | Multi-tier signoff (Draft → Moderation → Published) prevents erroneous mark release. |
| **ADR-017** | Lifetime Student Transcripts | FUNC | Implemented | Permanent access to digitally signed PDF transcripts and report cards for alumni. |
| **ADR-018** | Presigned S3 Direct Uploads | NFR | Implemented | 60s HMAC presigned direct client uploads; 0MB server RAM and 0% CPU load. |
| **ADR-019** | Strict Idempotency Key Engine | NFR | Implemented | Atomic Redis lock ( `SET NX EX 60` ) prevents duplicate payment entries on mobile. |

| ID | DECISION TITLE | CATEGORY | STATUS | CORE ARCHITECTURAL RATIONALE & OPERATIONAL IMPACT |
|---|---|---|---|---|
| **ADR-020** | Read-Heavy Upstash Redis Cache | NFR | Implemented | >90% hit ratio for master calendars and subjects; offloads 80% DB read load. |
| **ADR-021** | Inngest Headless Async Worker | NFR | Implemented | Buzzer Token pattern prevents 504 timeouts during 500+ batch PDF compilation. |
| **ADR-022** | Mobile-First Responsive Design | UX | Implemented | Fluid auto-fit grids and table scrolling for flawless 390px–500px mobile viewing. |
| **ADR-023** | Role-Adaptive Clean Dashboards | UX | Implemented | Eliminates cognitive fatigue; role-specific UI for Teachers, Bursars, and Parents. |
| **ADR-024** | Institutional Design Token System | UX | Implemented | Distinctive Burgundy/Cream/Bronze palette projecting institutional authority. |